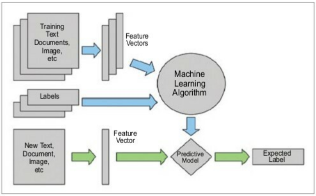

Fig. 2: Supervised learning model (Image courtesy: Google images)

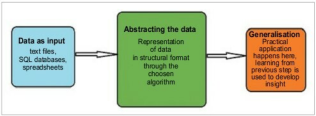

Fig. 3: Steps to implement machine learning

Reinforcement learning. Here, the system is given two different sets of input data and it needs to implement reinforcement machine learning technique to learn and identify the general pattern or hypothesis in one of the given set of inputs.

There can be more than one hypothesis derived but, finally, the optimal one is used by the system to derive the output for the other set of inputs. This is like learning the rules of a game by playing against an opponent.

Implementing machine learning in real life

Let us now look at implementing machine learning in real-life scenarios. You need to check how you can teach machines to take decisions and do your work just as you would do it by applying your own sense or logic.

In the course of teaching machines, every stage of the process helps to build a better version of the machine. There are five basic steps that need to be followed prior to letting a machine perform any unsupervised task.

Collecting data. This is one of the first and foremost steps in implementing any type of machine learning technique. Data plays a significant role in machine learning, whether it is in the form of raw data from MS Excel, Access or even text files. This step lays the foundation of future learning. We must be aware of the fact that the better the variety, volume and density of relevant data, the better will be the learning prospects for the machine.

Preparing data. Once data is collected, check the quality of what will be fed as training data to the system. You need to spend time in order to determine the quality of data and, accordingly, take steps to fix issues such as treat-